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EFFECTS OF TASK DIFFICULTY AND INTERUNIT INTERDEPENDENCE ON INFORMATION PROCESSING SYSTEMS

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The roles of task uncertainty within subunits and interunit interdependence in explaining variations in organizational designs have been subject to much theorizing (Fry, 1982; Gerwin, 1981; McCann & Galbraith, 1981). Previous research has found task uncertainty in subunits to be related (1) to personal means of intraunit coordination (Hrebiniak, 1974; Mohr, 1971; Pennings, 1975a; Tushman, 1979a; Van de Ven, Delbecq, & Koenig, 1976); (2) to boundary-spanning activity (Leifer & Huber, 1977; Tushman, 1977); and (3) to such facets of intraunit structuring as participation in decision making (Mohr, 1971) and autonomy in task performance (Schoonhoven, 1981). Perrow (1967) identified two components of task uncertainty—number of exceptions and analyzability—that are conceptually distinct but empirically related (Withey, Daft, & Cooper, 1983). The present study focuses on the analyzability component of task uncertainty, which we labeled task difficulty. Task difficulty refers to knowledge of cause-effect relationships (Van de Ven & Delbecq, 1974). Part of this study relates task difficulty to boundary-spanning activity, participation, and autonomy.

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Little empirical work on the relationship between interdependence and coordination mechanisms exists (Cheng, 1983). Only Tushman (1977, 1978) studied linkage between interunit interdependence and boundary-spanning activity; these studies concerned relationships between laboratory units and other units in a high technology firm. The present study also relates interdependence between units to interunit boundary-spanning activity in the context of an administrative system; it therefore complements Tushman's research.

No study has specifically addressed the relationship between boundary-spanning activity, one mechanism that coordinates relations between units, and elements of intraunit structuring. Research addressing these linkages has either combined forms of boundary-spanning activity or has used global measures of interaction (Aiken & Hage, 1968; Bacharach & Aiken, 1977; Hrebiniak, 1974; Leifer & Huber, 1977; Mohr, 1971). Finally, this study relates boundary-spanning activity to participation in decision making and to autonomy.

THEORETICAL ANALYSIS AND HYPOTHESES

Information processing, defined as "the gathering, interpreting, and synthesis of information in the context of organizational decision-making" (Tushman & Nadler, 1978: 614), has served to explain variations in organization designs. The capacity to process information is a function of a unit's design that includes (1) mechanisms linking units and (2) intraunit structuring. Task difficulty and interunit interdependence are two factors that determine requirements for information processing capacity. Because requirements and capacity should match, these factors are linked to the designs of work units (Galbraith, 1973, 1977; Tushman & Nadler, 1978).

Boundary Spanning Activities

Organizational subunits manage their interunit interdependencies through mechanisms that vary in information-processing capacity (McCann & Ferry, 1979; Thompson, 1967). Programming, although the least costly mechanism, can only deal with predictable events in which cause-effect relations are clear (Connolly, 1977; Galbraith, 1972). Supervisors, acting as boundary spanners, may deal with interunit decisions not falling within procedures. An alternative mechanism, direct lateral contact among subordinates, may provide less control and be more costly than supervisory boundary spanning (Galbraith, 1972). Coding or linguistic barriers (Tushman, 1979a) may also hinder contact among subordinates. However, if information-processing requirements exceed supervisors' capacities, organizations use supplementary mechanisms (Thompson, 1967; Tushman & Nadler, 1978; Van de Ven et al., 1976). These mechanisms include boundary spanning by unit members through gatekeepers, in instances in which differentiation is high (Tushman, 1977; Tushman & Scanlan, 1981), and boundary spanning through direct contact where such barriers are low (Leifer & Huber, 1977).

Such transactions may range from one-on-one meetings to joint problem-solving teams (Van de Ven et al., 1976) and may be assisted by third parties (Lawrence & Lorsch, 1967).

McCann and Ferry stated, "Interdependence exists when actions taken by one referent system affects [sic] the actions or outcomes of another referent system" (1979: 113). In this paper, our concern was the degree of policy and procedural interdependence between focal units and other units as indicated by changes that other units induced in focal units.¹ The greater the interdependence, the greater would be the need to gather, analyze, and distribute information between subunits. Boundary-spanning activity by unit members, supplementing the limited capacity of rules and hierarchy (Tushman, 1977, 1978), may meet this need for information processing. Therefore, we expect that:

Hypothesis 1: The greater the degree of interunit interdependence, the greater the amount of boundary-spanning activity by unit members.

It is a central theme in previous research that the greater the task difficulty, the greater the requirement for information-processing capacity (Duncan, 1973; Galbraith, 1977; Tushman & Nadler, 1978). Lateral communications that supplement programming and hierarchy in providing advice and information to unit members in task performance may meet such requirements (Burns & Stalker, 1961). These networks would provide (1) reduced equivocality of information (Daft & Macintosh, 1981); (2) conceptual frameworks (Connolly, 1977); (3) social validation of decisions (Connolly, 1977; Litwak, 1961); and (4) coordination (March & Simon, 1958; Thompson, 1967).

When interunit interdependence is low, these networks may be restricted to intraunit transactions, with supervisors managing interfaces between units. However, with high levels of interdependence, expanding networks to include members from other units may facilitate needed extraunit information exchange (Galbraith, 1972). Therefore, since the level of interunit interdependence for most of the work units studied was moderate or high, we propose that:

Hypothesis 2: The greater the degree of task difficulty, the greater the amount of boundary-spanning activity by unit members.

Intraunit Structuring

Previous research has developed and tested linkages between task difficulty and participation and has found varying levels of support (Hage & Aiken, 1969; Hrebiniak, 1974; Grimes, Klein, & Schull, 1972; Mohr, 1971). Briefly, participation by subordinates enhances supervisory capacity to generate and evaluate alternatives (Duncan, 1973). The more difficult the tasks, the greater the vesting of cause-effect knowledge in subordinates, hence the

¹ For other frameworks, see McCann and Ferry (1979); Pennings (1975b); Thompson (1967); and Van de Ven et al. (1976).

greater their potential contributions to decision making. Therefore, we hypothesize that:

Hypothesis 3: The greater the degree of task difficulty, the greater the level of participation in decision making by unit members.

The relationship between task difficulty and discretion at the point of task execution enjoys considerable theoretical and empirical support (Bell, 1967; Burns & Stalker, 1961; Grimes et al., 1972; Hrebiniak, 1974; Schoonhoven, 1981; Thompson & Bates, 1957). When task difficulty is low, subordinates may refer exceptions upward; but as cause-effect relationships become less tangible, units may expand information processing capacity by increasing discretion at lower levels. Granting authority to subordinates to make decisions within prescribed boundaries serves at least three functions: (1) it prevents overloading supervisors (Katz & Kahn, 1978); (2) it improves timeliness of responses; and (3) it places decisions at levels at which there may be greater expertise about particular issues than supervisors have. Control of subordinates may be exercised through professionalization, the inculcation of norms and values, or the specification of goals (Galbraith, 1973). Therefore, we expect that:

Hypothesis 4: The greater the degree of task difficulty, the greater the degree of autonomy of unit members.

Boundary-spanning activities imply both joint decision making with members of other units and needs to seek out and consolidate extraunit information in performing daily tasks. Thus levels of boundary-spanning activity are likely to be associated with the extent of subordinates' knowledge regarding other units. Where extraunit knowledge is dispersed, rather than concentrated among supervisors, there are potential subordinate roles in making decisions having extraunit considerations in at least two situations. First, supervisors would be partly dependent on subordinates for information and advice when making decisions with extraunit considerations. Although they differ in the form of boundary-spanning activity investigated, several studies (Aiken & Hage, 1968; Hrebiniak, 1974; Leifer & Huber, 1977; Mohr, 1971; Tushman, 1979a) have supported its linkage to participation. Second, such joint decision making and subordinate expertise in extraunit matters would be consistent with limited supervision in day-to-day decisions with extraunit considerations. The work of Aiken and Hage (1968), Bacharach and Aiken (1977), and Leifer and Huber (1977) supported this linkage. Therefore, we expect that:

Hypothesis 5: The greater the degree of boundary-spanning activity, the greater the level of unit members' participation in decision making.

Hypothesis 6: The greater the degree of boundary-spanning activity, the greater the degree of unit members' autonomy.

METHODS

Research Setting and Sample

The senior author carried out this study in the field and head office units of the Department of Employment Security of a western state as an unpaid researcher. Van de Ven and Delbecq (1974) described this agency's functions and structure. Following Van de Ven and colleagues (1976) and Tushman (1977, 1978), we used work groups as the units of analysis. First-line supervisors with at least one year as unit heads who had full-time subordinates with sufficient experience to be familiar with their positions answered questions on the subordinates' jobs. The study included all units meeting the criterion for supervisory experience and containing at least two experienced subordinates. Each potential respondent received an individually addressed questionnaire through the internal mail system. Respondents mailed completed questionnaires directly to the senior author. The data analysis includes only units in which responses came from both the supervisor and from at least two subordinates representing at least one-third of those qualified to respond. Questionnaires went out to 612 potential respondents in 131 work units; 381 questionnaires (62%) were returned. Seventy-five units met the criteria for data analysis; these units included 360 potential respondents of whom 283 responded for a 79 percent response rate.

Variables

Brief descriptions of each of the variables follow.² All items were measured on 7-point Likert scales. Factor analysis with varimax rotation was used to group items into variables. The items selected loaded at $> .40$ on one factor. Table 2 gives means, standard deviations, and reliabilities.

The four task difficulty items followed Perrow's (1967) concept of analyzability (e.g., the extent to which individuals use standard methods). The two interunit interdependence items concern change in policies and procedures induced by other subunits, a concept that applies to both $B \rightarrow A$ and $A \rightarrow B$ work flow sequences. An item measuring this concept's complement, change induced in other subunits, confounded induced change and interaction; hence the analysis omits it. However, that item's correlation with interdependence was .58 as compared with an interitem correlation for interdependence of .64. Thus, we considered the measure to indicate general levels of policy and procedural interdependence. The four boundary-spanning activity items measured the frequency and time spent in interactions with members of other units.

Frequency of participation was measured for each of four decisions on a scale ranging from almost always to almost never. Two of the decisions were task-related and two concerned personnel. A fifth item concerned influence in changes in methods initiated by subordinates. Autonomy was a composite of two items reflecting independence of action generally and two items refer-

² Full texts of items and data for all analyses are available from the senior author.

ring to independence in handling nonroutine decisions. Two of these specifically referred to independence from "supervisors or specialists" in making decisions. The measure confounded two aspects of autonomy: supervisory constraints and constraints placed upon members of focal units by other units. However, as Table 2 shows, interunit interdependence was not associated with autonomy ($r = -.08$), implying a bias toward supervisory constraints.

Pooling

To compute scores for work groups, supervisors' scores were added to the means of subordinates' scores, and these sums were divided by two (Van de Ven et al., 1976). Three tests supported this procedure. One-way analysis of variance (Dixon & Massey, 1969) supported pooling subordinate responses into work groups for all of the variables ($p < .005$). No variable violated the homogeneity of variance assumption.³ We dropped four units whose variances exceeded the pooled variance ($p < .05$) for two or more variables from the study.

Because there were scores for both supervisors and subordinates on each variable, a multitrait multimethod (MTMM) analysis (Campbell & Fiske, 1959) was conducted; it supported pooling supervisor and subordinate scores. First, the significance of the supervisor-subordinate correlations for each variable met the convergent validity criterion (Table 1). Second, if supervisor and subordinate scores were to be pooled, the pattern of relationships amongst the variables had to be similar in both heterotrait, monomethod blocks—the supervisor block and the subordinate block. Following Brief and Aldag (1978), we used Spearman rank order correlation to test the similarity of the two patterns. This yielded a correlation of .72 ($p < .05$) that supported pooling.

Methods Variance

Two tests assessed divergent validity. First, all 40 comparisons met the criterion that the convergent validity correlations ($r_{ii'}$) be greater than $r_{ij'}$ and $r_{i'j}$, where i and j denote supervisors' scores on variables i and j ; similarly, i' and j' denote subordinates' scores on variables i and j . The second criterion, that $r_{ii'}$ be greater than r_{ij} and $r_{i'j'}$, was met in 31 of the 40 comparisons. Table 1 shows the nine exceptions. Interdependence and boundary-spanning activity show differing patterns with other variables (Table 2), thus mitigating concern about common methods variance. However, conclusions regarding autonomy bear a caveat on potential methods variance.

RESULTS AND DISCUSSION

Hypotheses 1 and 2 stated that interunit interdependence and task difficulty would be positively related to boundary-spanning activity. Results supported these hypotheses with $p < .001$ and $p < .01$, respectively (Table 2).

³ Lowest $p = .30$ with the Bartlett - Box test.

TABLE 1
Results of Convergent and Divergent Validity Analyses^a

Variables	Convergent Validity	Exceptions to Divergent Validity Criterion	
	Supervisor-Subordinate Correlations ($r_{ij'}$)	Supervisor Correlations ($r_{ij'} < r_{ij}$)	Subordinate Correlations ($r_{ij'} < r_{i'j'}$)
1. Task difficulty	.58***		
2. Interunit interdependence	.30**	$r_{23} = .52***$	$r_{2'3'} = .50***$
3. Boundary-spanning activity	.50***	$r_{32} = .52***$	$r_{3'2'} = .50***$
4. Participation in decision making	.48***		
5. Autonomy	.28**	$r_{51} = .47***$ $r_{53} = .34**$ $r_{54} = .32**$	$r_{5'1'} = .32**$ $r_{5'4'} = .34**$

TABLE 2
Descriptive Statistics and Zero-Order Correlations among All Variables

Variables	Unit Means ^a	Standard Deviations	Reliabilities ^b	Zero-Order Correlations among Variables ^a				
				2	3	4	5	
1. Task difficulty	3.64	0.98	0.72	.03	.32**	.21*	.44***	
2. Interunit interdependence	3.82	1.18	0.68		.51***	.09	-.08	
3. Boundary-spanning activity	3.59	0.94	0.80			.38***	.30**	
4. Participation in decision making	4.21	1.04	0.79				.28*	
5. Autonomy	4.69	0.81	0.71					

^a $N = 71$

^b α ; $N = 269$

* $p < .05$

** $p < .01$

*** $p < .001$

These findings from an administrative setting complement Tushman's (1977, 1978) findings in a high technology setting. However, although Tushman found boundary-spanning activity to be concentrated in a few roles, the data in the present study did not permit an analysis of its distribution.

Hypothesis 3, that task difficulty would be positively related to participation, received only weak support ($r = .21, p < .05$), which is consistent with Mohr's (1971) and Hrebiniak's (1974) results. When task difficulty was correlated with each of the decisions, only two correlations were significant at $p < .05$; these were number of people and influence in subordinate-initiated changes in methods. Task difficulty was not related to two task decisions, introduction of new tasks and supervisor-initiated changes in methods. Other considerations, such as a need for acceptance of decisions (Vroom & Yetton, 1973), may overshadow task difficulty in predicting participation (Mohr, 1971). However, Duncan (1973) showed that linkage between uncertainty and participation was contingent upon the nonroutineness of decisions themselves, a characteristic this index did not measure. Therefore, this finding may reflect a flaw in our methods.

Hypothesis 4, linking task difficulty with autonomy, was supported at $p < .001$, consistent with studies cited earlier in this paper (e.g., Schoonhoven, 1981). In this study, autonomy was the dependent variable; however, Glisson (1978) argued for a linkage from autonomy to task difficulty, suggesting that workers given discretion may perceive variations in clients' needs and undertake nonroutine searches to solve problems. Nevertheless, in contrast to findings for participation, these results support Mohr's (1971) position that autonomy and participation be considered as separate variables, rather than be subsumed under a rubric such as "organicness" (Leifer & Huber, 1977:237). In this light, these variables make different contributions to information-processing capacity; therefore they may enjoy different relationships with variables that affect information-processing requirements.

Hypotheses 5 and 6, stating that boundary-spanning activity would be positively linked with participation and autonomy, received support at $p < .001$ and $p < .01$, respectively; these results were consistent with Leifer and Huber (1977). One contribution of this paper was isolating interunit boundary-spanning activity as affecting two dimensions of subunit structure, both of which are aspects of the influence of subordinates in decision making. Although our rationale included participation and autonomy as dependent variables, the relationships may be reversed or there may be mutual causation. Participation in decisions with extraunit implications may require boundary-spanning activity during implementation; where autonomy and interunit interdependence exist, unit members may be encouraged to seek out extraunit information through personal efforts. Longitudinal research to assess causal directions would facilitate understanding how changing designs to match requirements and capacity for information processing may stimulate shifts, or pressures for shifts, in the degree of influence subordinates exercise in decision making. Thus, such research would have implications for supervisory

style. It would also help to reveal how shifts in subordinate influence may cause design changes to emerge.

This study has a number of limitations. First, as the Methods section indicated, there was potential common methods variance for autonomy. Second, only one organization, a people-processing state agency, was studied, and the units of analysis were work groups whose members had no supervisory responsibilities. Finally, this study used no measures of effectiveness; consequently we can draw no conclusions about whether the relationships found characterized effective or ineffective units.

CONCLUSIONS

Results support all six hypotheses and also support and extend research in information processing as an approach to understanding organizational design and structure. Two particular themes may contribute to further research in this area. The first is the need to specify design elements' particular contributions to information-processing capacity and relate these to specific demands for capacity that such factors as task difficulty induce. The second is the relationship between coordination mechanisms and elements of intraunit structure. Research in this area may assist practitioners by helping to predict changes in relationships between supervisors and subordinates that design changes may induce—or vice versa.

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SOLUTION DIVERSITY AND THE QUALITY OF GROUP DECISIONS

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Research investigating the effects of diversity on group decision making has been extensive. For example, Shaw's (1981) review discussed the effects

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